Aaryan Patel

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PROFILE

Third-year University of Toronto student-athlete majoring in Economics with Data Analytics (minors: Computer Science, Statistics). Experience spans data science internship work in experimentation, analytics, and applied machine learning, alongside product strategy and agentic AI engineering on an independent venture. Proven ability to perform under pressure and manage competing demands, balancing varsity athletics with a demanding academic program.

EDUCATION

University of Toronto | Economics with Data Analytics | Minors: CS & Statistics 2024 – Present
Relevant courses: Advanced Microeconomic Theory, Advanced Macroeconomic Theory, Probability & Statistics I, Linear Algebra, Big-Data Tools for Economists, Foundations of CS I & II, Software Design.

Varsity Student-Athlete (Rowing).

Menlo Atherton High School | High School Diploma 2020 – 2023
Founded MA Hackers Club (2023); completed AP CS A & Principles and college-level Economics.

PROFESSIONAL EXPERIENCE

Product Manager | DeUNA September 2026 – Present
Payments Orchestration Platform Connecting Major F100 Companies, Tech, Airlines, and QSRs Across NA, LATAM, and EU

Transitioned from Data Science Intern to Associate Product Manager of the ML Optimizer for Merchant-Initiated Transactions and its related agentic workflows.

Continuing to contribute part-time during the school year alongside a full course load.

Data Science Intern | DeUNA July 2026 – September 2026
Payments Orchestration Platform Connecting Major F100 Companies, Tech, Airlines, and QSRs Across NA, LATAM, and EU

Led A/B testing for payment request optimization end to end, designing the experiment structure, defining KPIs and success metrics, and identifying feature opportunities through data analysis; projected to increase annualized GMV for DeUNA merchants by $230M.

Delivered actionable insights from SQL-driven analysis across DeUNA's four-layer payments architecture (Snowflake, Postgres, Redis), increasing GMV by over $300K annually.

Built and trained MVP reconciliation models combining a gradient boosting ranker for transaction matching with a neural network schema detector for PSP settlement drift, with performance strong enough to drive customer adoption of the product and win executive buy-in for additional staffing and funding.

Conducted approval rate analysis, fraud model evaluation, and decline code normalization across the payments pipeline to support product and risk decisions.

Implementation Engineering Intern | Strand Analytica October 2026 – Present
AI Wargaming Simulations

Delivering custom-built harnesses and AI layers for the British Ministry of Defence and the U.S. Department of War, by building tailored wargaming solutions that inform strategic decision-making in wartime environments.

Continuing to contribute part-time during the school year alongside a full course load.

AI Product Intern | LookAcross AI May 2026 – July 2026
AI-Native GTM Intelligence Startup

Improved organizational efficiency for the founder by surfacing real-time insights on priorities and activity across the org, by implementing agentic workflows that connected 8+ tools (Gmail, Slack, Linear, Fireflies, Calendar, Drive) into a single unified dashboard, built with security and accuracy as first-class design constraints using the Anthropic Claude API, OpenRouter for multi-model orchestration, Model Context Protocol (MCP) for structured tool access, and retrieval-augmented generation for grounding insights in source data.

Moved an event intelligence feature from concept to active development, covering problem framing, competitive analysis, data model design, phased delivery estimates, and risk assessment, by owning product definition and technical scoping end to end.

Built the technical foundation for the event intelligence pipeline, spanning agentic scraping workflows, entity resolution logic to match scraped attendees against partner records, and white-space scoring integrated with the existing Partner Intelligence Engine, by implementing the system architecture alongside the product specification.

Engineering Intern | American Precision Gear Summer 2024
DoD CNC Contract Shop

Maintained quality standards on critical aircraft and space vehicle parts across a high-volume production floor processing 1,000–2,000 parts per week, by executing QC and inspection processes, and surfaced process inefficiencies for shop floor stakeholders by cleaning and analyzing messy QC datasets (1,000+ inspection records).

Built hands-on manufacturing proficiency operating mills, routers, and cutters on the shop floor, while reducing manual overhead across inspection, logistics, and documentation workflows by leveraging AI tools and managing secure, on-time delivery of finished components.

RESEARCH EXPERIENCE & PROJECTS

Independent Project | Athenaeum — AI-Assisted Storytelling Platform for Indie Filmmakers | Feb – July 2026
Owned product strategy and roadmap for Athenaeum, a platform that turns filmmakers' voice memos and personal writing into industry-ready treatments, translating design-partner feedback into prioritized specs and PRDs across a 6-layer product architecture.

Built the backend (FastAPI, Postgres, pgvector) for a RAG-powered voice preservation system that extracts a filmmaker's stylistic fingerprint from their writing via Verbalized Sampling and generates treatments in industry formats (Sundance, IDA, ITVS), shipping the MVP from concept to a working build.

Engineered a Dramaturg agent that asks clarifying questions and surfaces narrative gaps rather than auto-generating plot, paired with an authorship ledger tracking the provenance of every output byte, central to the product's differentiation and trust strategy with filmmakers.

University of Toronto | Neighborhood Home Values & Police Search Outcomes — ECO225 | Spring 2026
Tested whether neighborhood home values predict the likelihood a police stop results in a search, by merging 250,093 SFPD stops with Zillow Home Value Index data at the neighborhood level.

Captured linear and interaction effects across subgroups, by building 8 OLS regression specifications with HC1 robust standard errors and producing choropleth maps and scatter plots of spatial enforcement patterns.

Captured non-linear enforcement patterns that linear specifications missed, by implementing regression tree and random forest models atop HTML-based web scraping (BeautifulSoup) for supplementary data, and cut ML pipeline development time by leveraging Claude Code for autonomous coding and debugging.

Found that home values predict search likelihood, largely mediated by low-level stop concentration, consistent with order-maintenance policing theory (Harcourt 2001, Beck 2020), by synthesizing results across all model specifications.

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, Scikit-learn, SciPy), SQL, MATLAB, Java, C++, R (basic), JavaScript (basic)

AI & LLM Development: Prompt Engineering, LLM API Integration (Anthropic Claude API, OpenAI GPT-4, OpenRouter), Agentic Coding (Claude Code), System Prompt Design, Multi-model Orchestration, RAG concepts

ML, Statistics & Causal Inference: OLS/IV, Logistic Regression, CART, Random Forest, Gradient Boosting, Neural Networks, Regularization (Ridge/Lasso), Clustering, Time Series, Cross-Validation, Hyperparameter Tuning, Feature Engineering, DiD, Panel Data, Spatial Econometrics, Robust SEs (HC1), Hypothesis Testing, A/B Testing

Cloud & Data Infrastructure: AWS (SageMaker, S3, Glue, Lambda, EC2), GCP, Snowflake, Spark/PySpark, dbt, ClearML, Postgres, Redis, REST APIs, Data Pipeline Construction, Large-Scale Dataset Management

APIs Used: Google Maps Places API, Spotify API, Anthropic API, BeautifulSoup (web scraping)

Visualization & Tools: Matplotlib, Plotly, Tableau, Choropleth Mapping, Jupyter, Git, VS Code, Excel/Google Sheets

LEADERSHIP & OUTREACH

Varsity Rower & Fundraising Director | UofT / Norcal Rowing Crew 2020 – Present
Balanced Division I-level athletic competition with a full UofT course load, captaining Norcal Crew (250+ athletes) to multiple national championships and earning selection to the U.S. Rowing Olympic Development Program; now manages stakeholder relationships across a national alumni network as Alumni and Fundraising Student Director.

Lead Coordinator & Guest Speaker | Community STEM Outreach 2023 – 2024
Bridged the digital divide for audiences of 50+, by designing and delivering computer literacy and coding curricula for underserved communities and seniors through mentorship.

AWARDS & RECOGNITION

College Board National African American & Hispanic Recognition Award (2024) • Norcal Rowing: Leadership Award (2024), Best Teammate (2023), Hardest Worker (2021)